Machine Learning

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Outline

- Generative Adversarial Networks (GANs)

Generative Models

- A generative model is a type of machine learning model that aims to learn the underlying patterns or **distributions of data** in order to **generate new, similar data**.

- In essence, it's like teaching a computer to dream up its own data based on what it has seen before.

- The significance of this model lies in its ability to create, which has vast implications in various fields, from art to science.

Generative vs. Discriminative Models

- **Generative models:** These models focus on understanding how the data is generated.

- They aim is to learn the distribution of the data itself.

- For instance, if we're looking at pictures of cats and dogs, a generative model would try to understand what makes a cat look like a cat and a dog look like a dog. It would then be able to generate new images that resemble either cats or dogs.

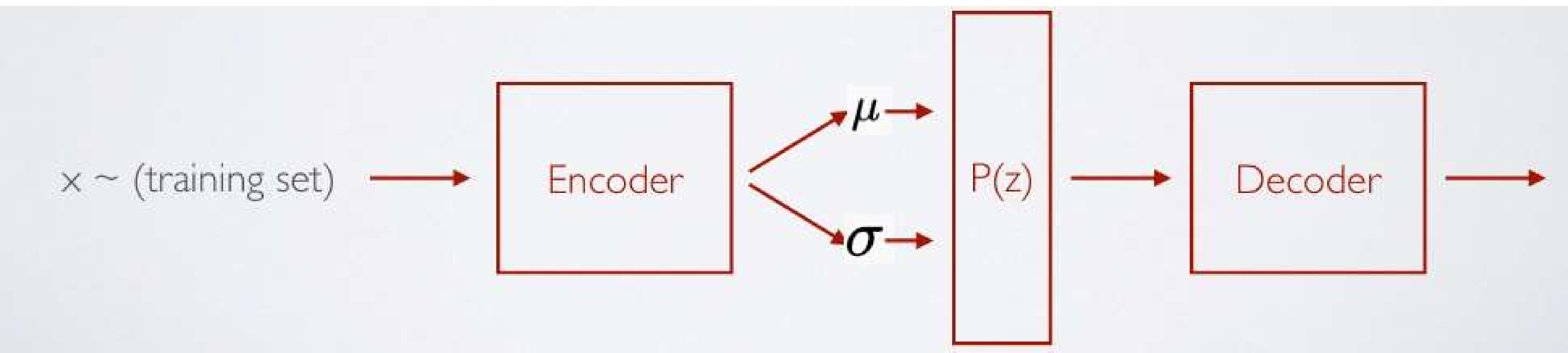
Generative vs. Discriminative Models

- **Discriminative models:** These models, on the other hand, focus on distinguishing between different types of data.

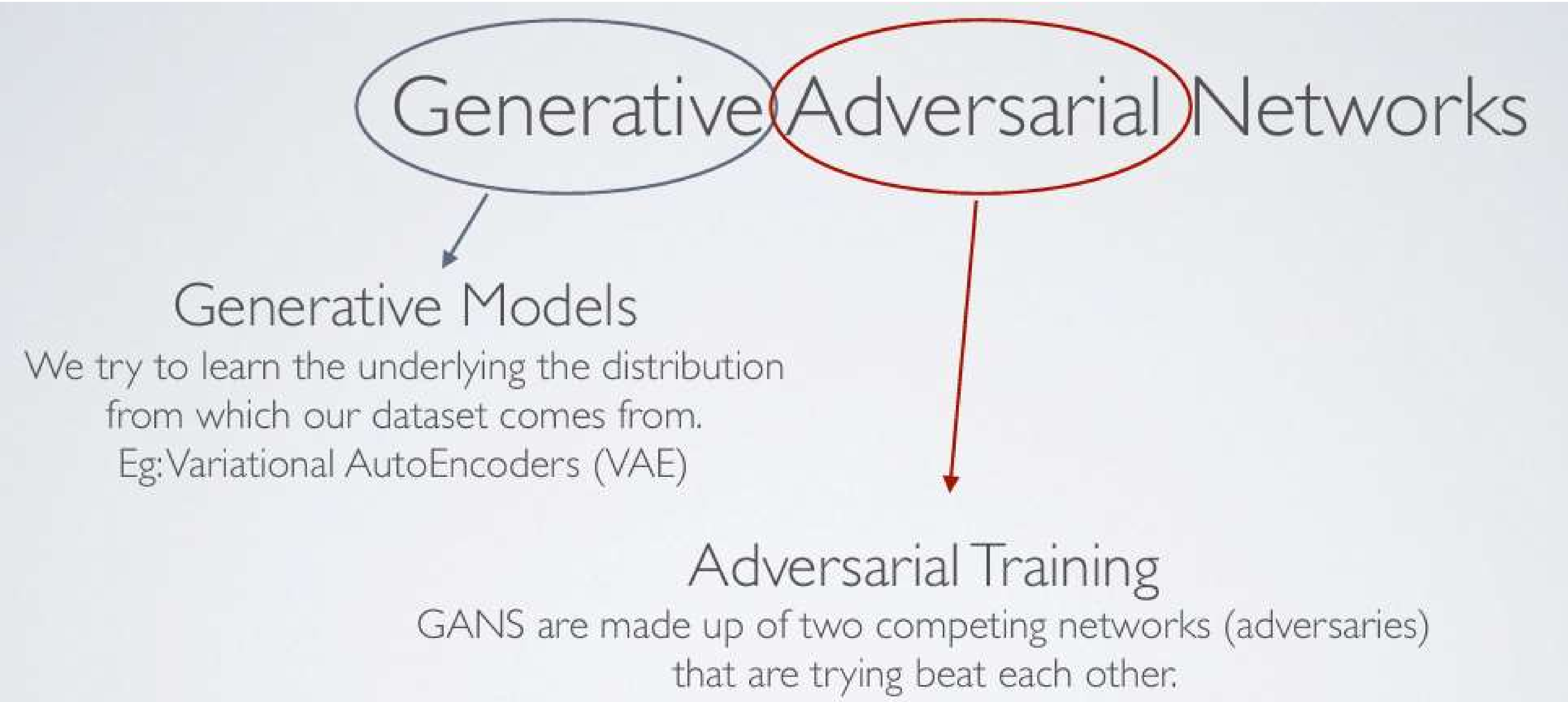
- They don't necessarily learn or understand how the data is generated; instead, they learn the boundaries that separate one class of data from another.

- Using the same example of cats and dogs, a discriminative model would learn to tell the difference between the two, but it wouldn't necessarily be able to generate a new image of a cat or dog on its own.

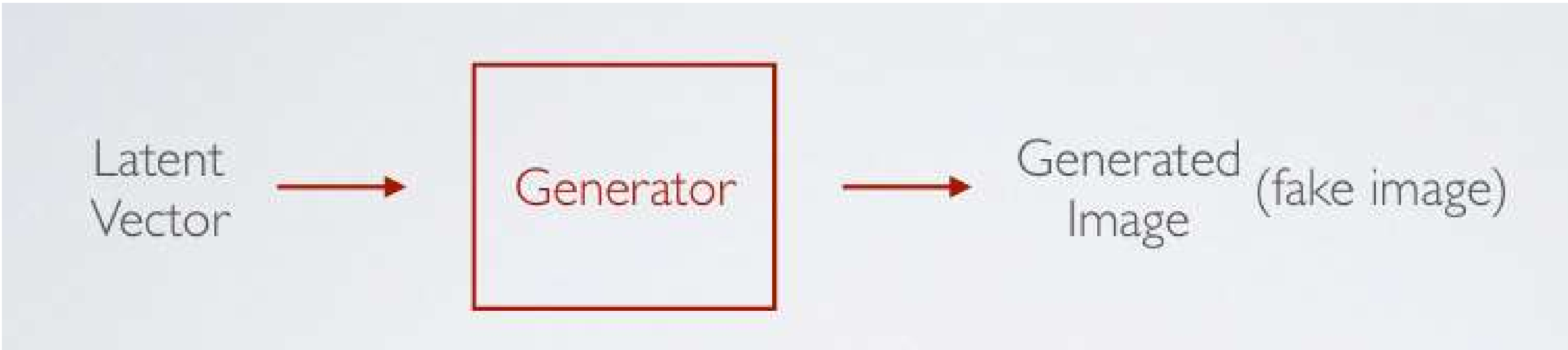
Generative Adversarial Networks



Generative Adversarial Networks



Generative Adversarial Networks



Generative Adversarial Networks

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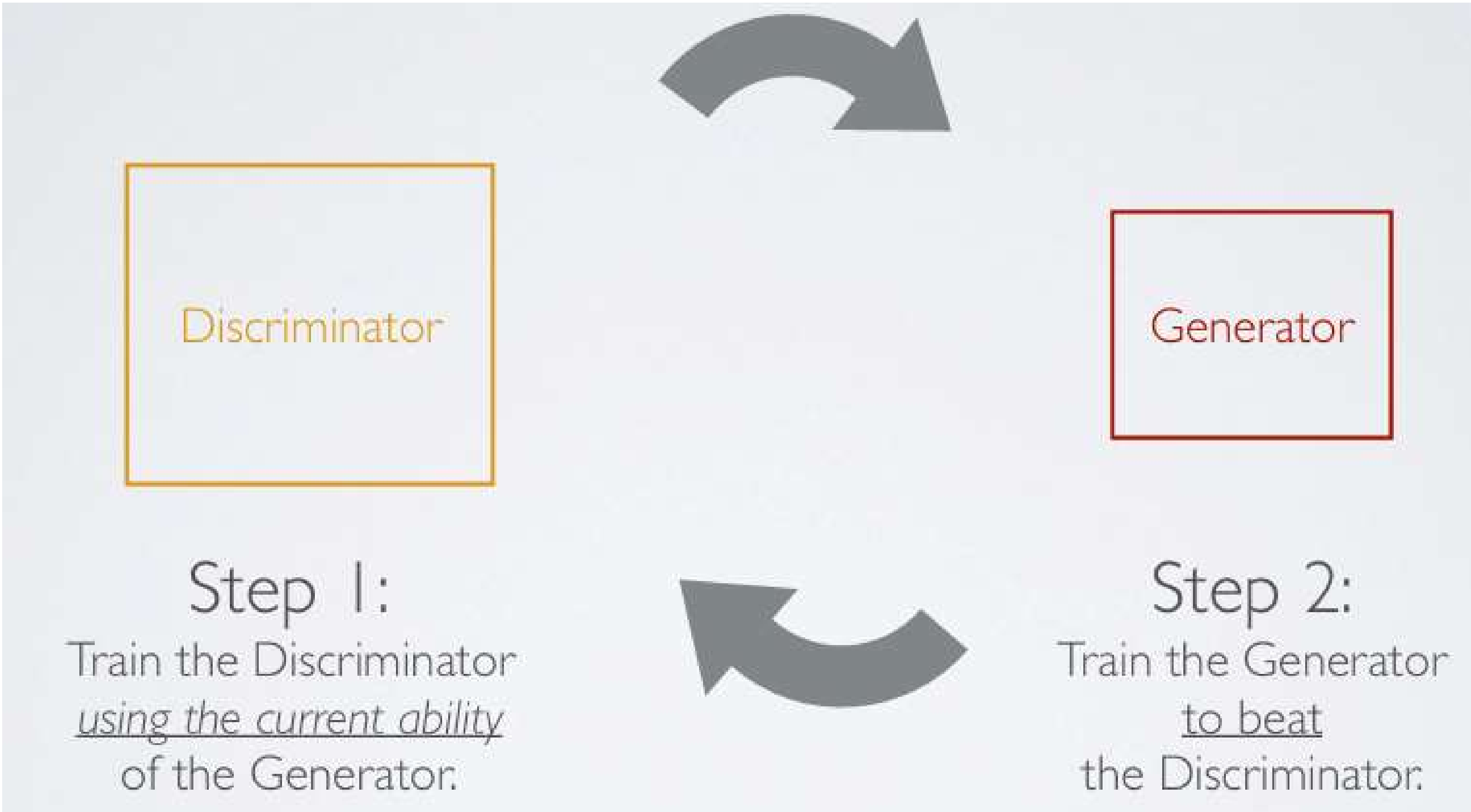
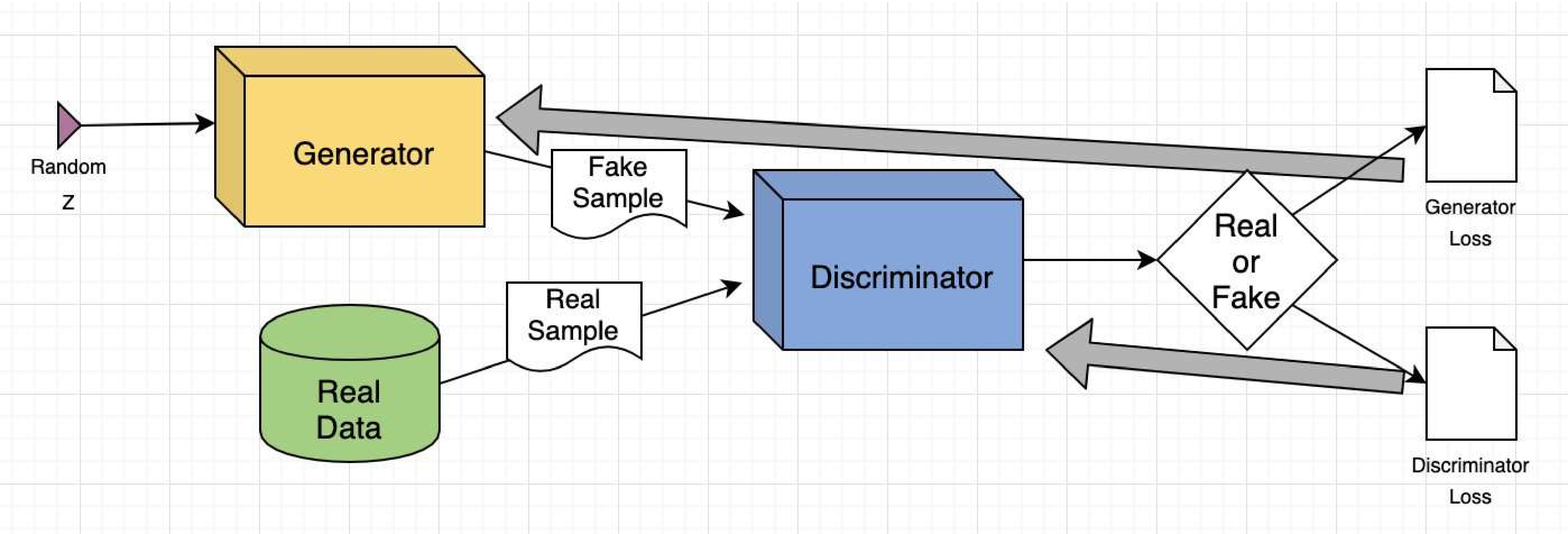
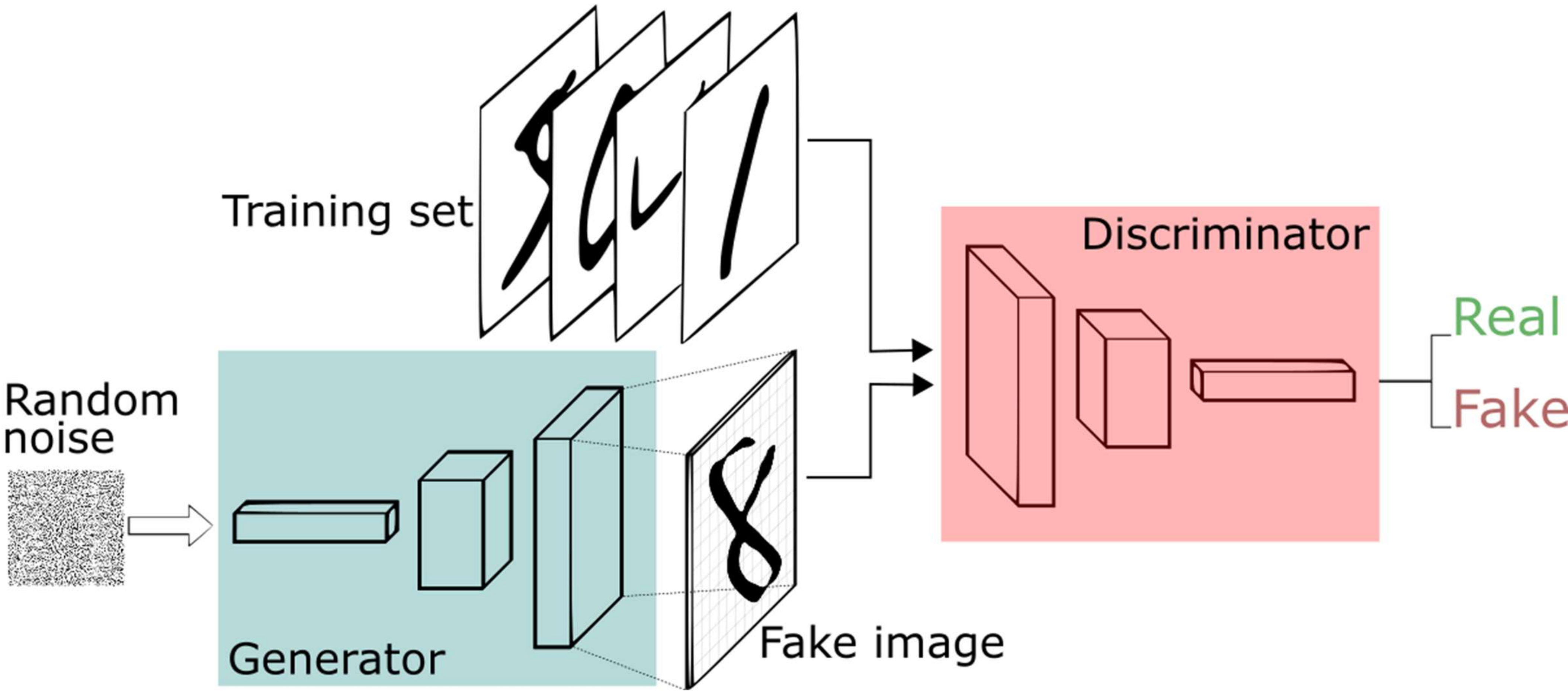


Generative Adversarial Networks



Generative Adversarial Networks

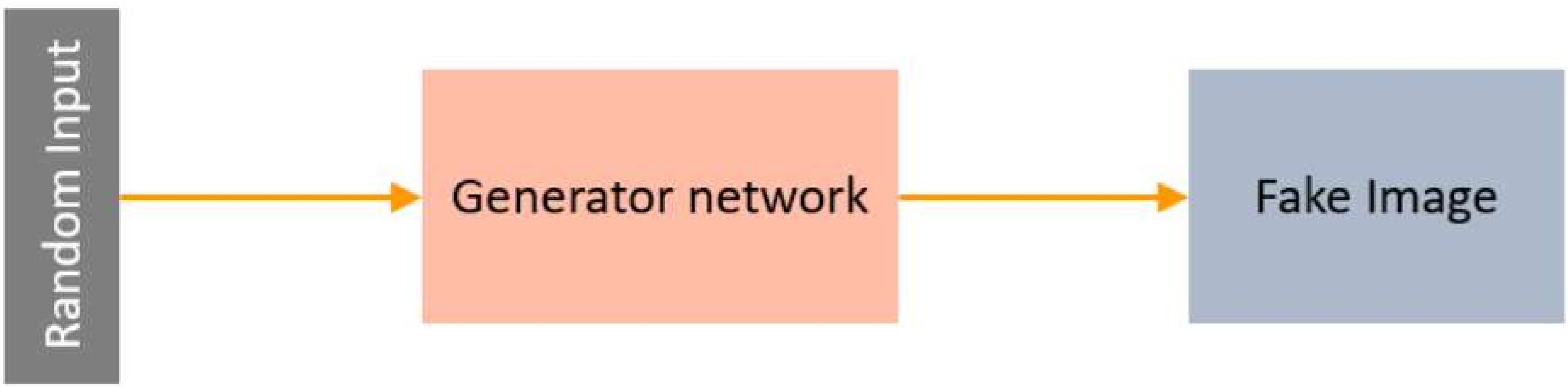


GANs Architecture

- GANs consists of two neural networks.
- There is a **Generator G(x)** and a **Discriminator D(x)**.
- Both of them play an adversarial game.
- The generator's aim is to fool the discriminator by producing data that are similar to those in the training set.
- The discriminator will try not to be fooled by identifying fake data from real data.
- Both of them work simultaneously to learn and train complex data like audio, video, or image files.

What is a Generator?

- A Generator in GANs is a neural network that creates fake data to be trained on the discriminator.
- It learns to generate reasonable data.
- The generated examples/instances become <u>negative training</u> examples for the discriminator. It takes a fixed-length random vector carrying noise as input and generates a sample.



What is a Generator?

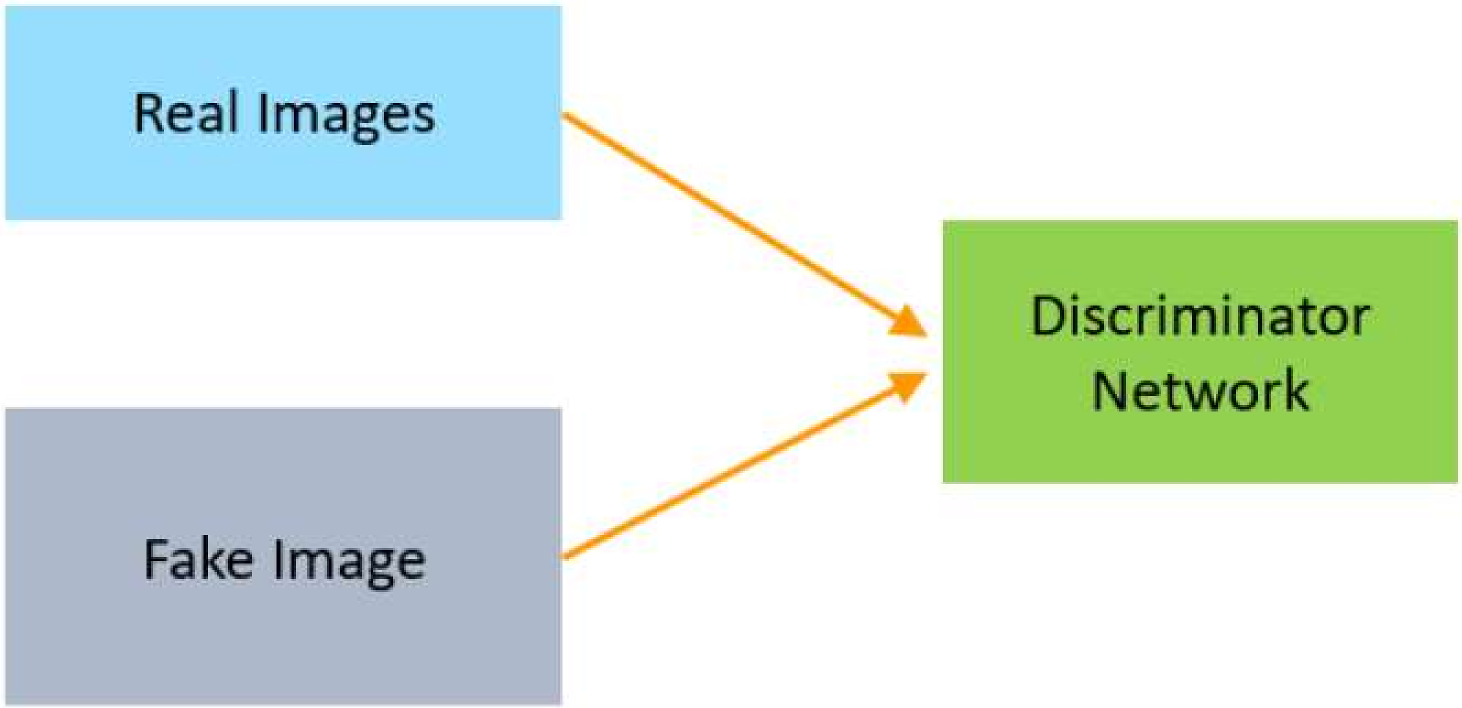
- The main aim of the Generator is to make the discriminator classify its output as real. The part of the GAN that trains the Generator includes:
- noisy input vector
- generator network, which transforms the random input into a data instance
- generator loss, which penalizes the Generator for failing to dolt the discriminator.
- The backpropagation method is used to adjust each weight in the right direction by calculating the weight's impact on the output. It is also used to obtain gradients and these gradients can help change the generator weights.

What is a Discriminator?

- The Discriminator is a neural network that identifies real data from the fake data created by the Generator. The discriminator's training data comes from different two sources:

- The real data instances, such as real pictures of birds, humans, currency notes, etc., are used by the Discriminator as positive samples during training.
- The fake data instances created by the Generator are used as negative examples during the training process.

What is a Discriminator?



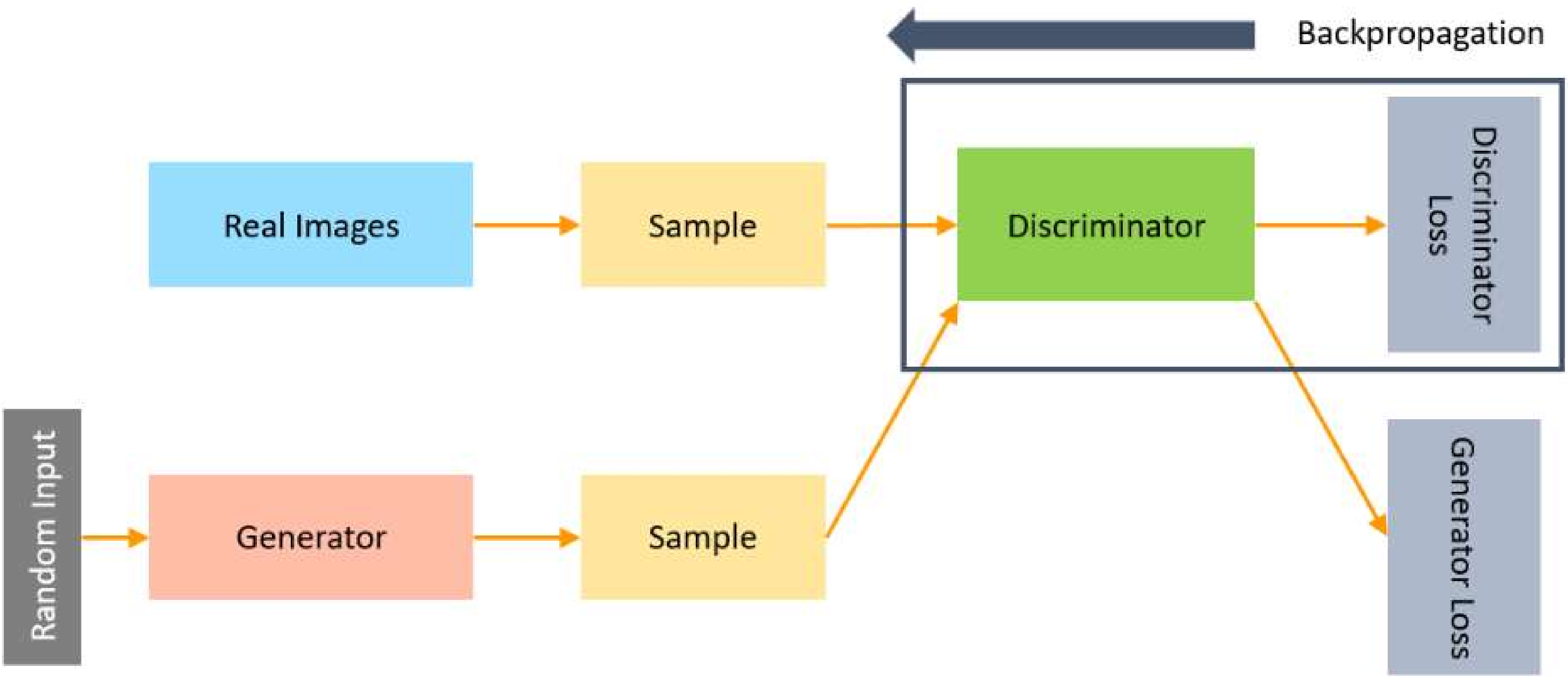
What is a Discriminator?

- While training the discriminator, it connects to two loss functions. During discriminator training, the discriminator ignores the generator loss and just uses the discriminator loss.

- In the process of training the discriminator, the discriminator classifies both real data and fake data from the generator. The discriminator loss penalizes the discriminator for misclassifying a real data instance as fake or a fake data instance as real.

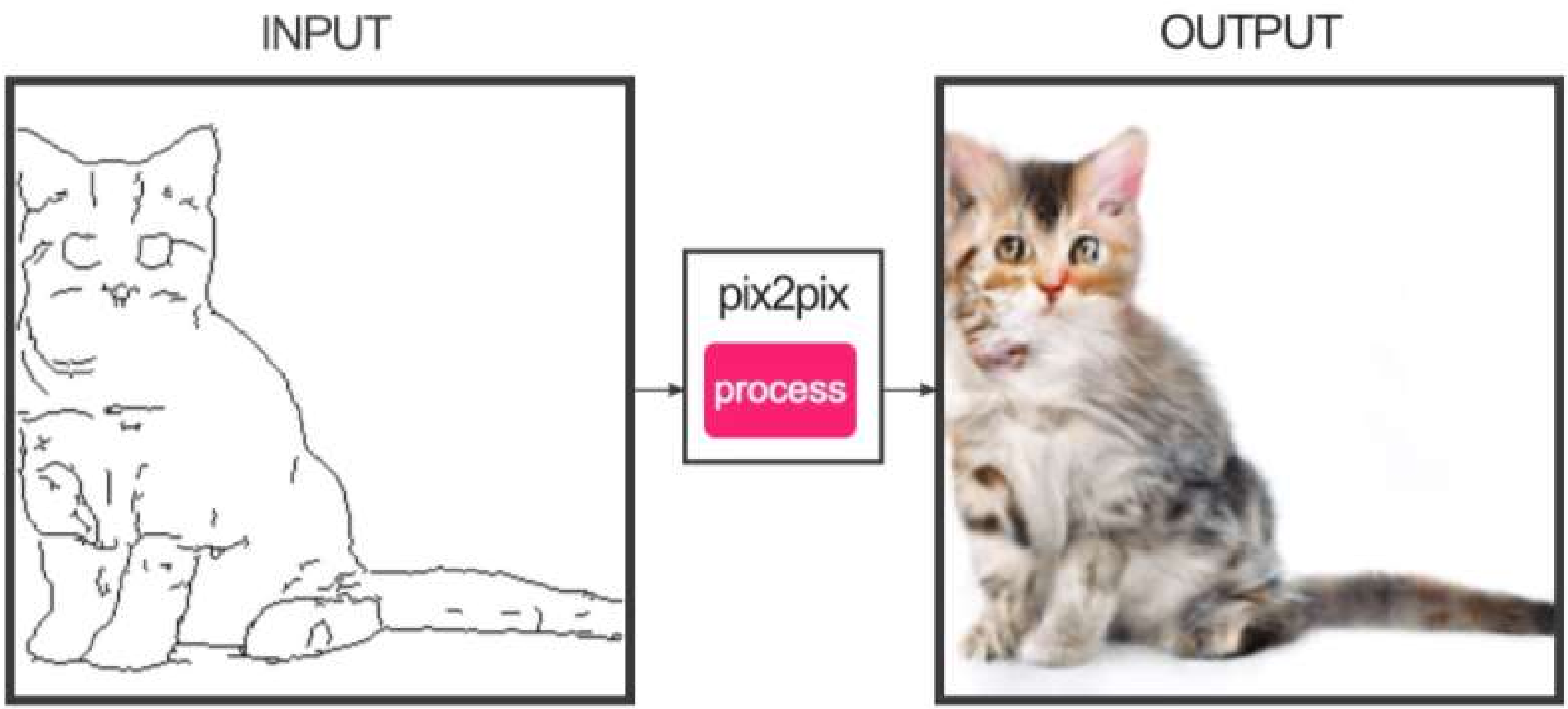
- The discriminator updates its weights through backpropagation from the discriminator loss through the discriminator network.

GANs Architecture



GANs Applications

- Image-to-image translation: Image-to-image translation is an application where a certain image B is transformed with the properties of A.



GANs Applications

- Generate Human Faces: GANs can be trained on the images of humans to generate realistic faces.

GANs Applications

- Clothing translation is converting an image of clothing from one style or design to another. GANs have been used to develop systems that can translate images of clothing from one type to another, such as changing the color or pattern of a shirt or dress.

GANs Applications

- Generate images based on input text.

Summary

- Generative Adversarial Networks